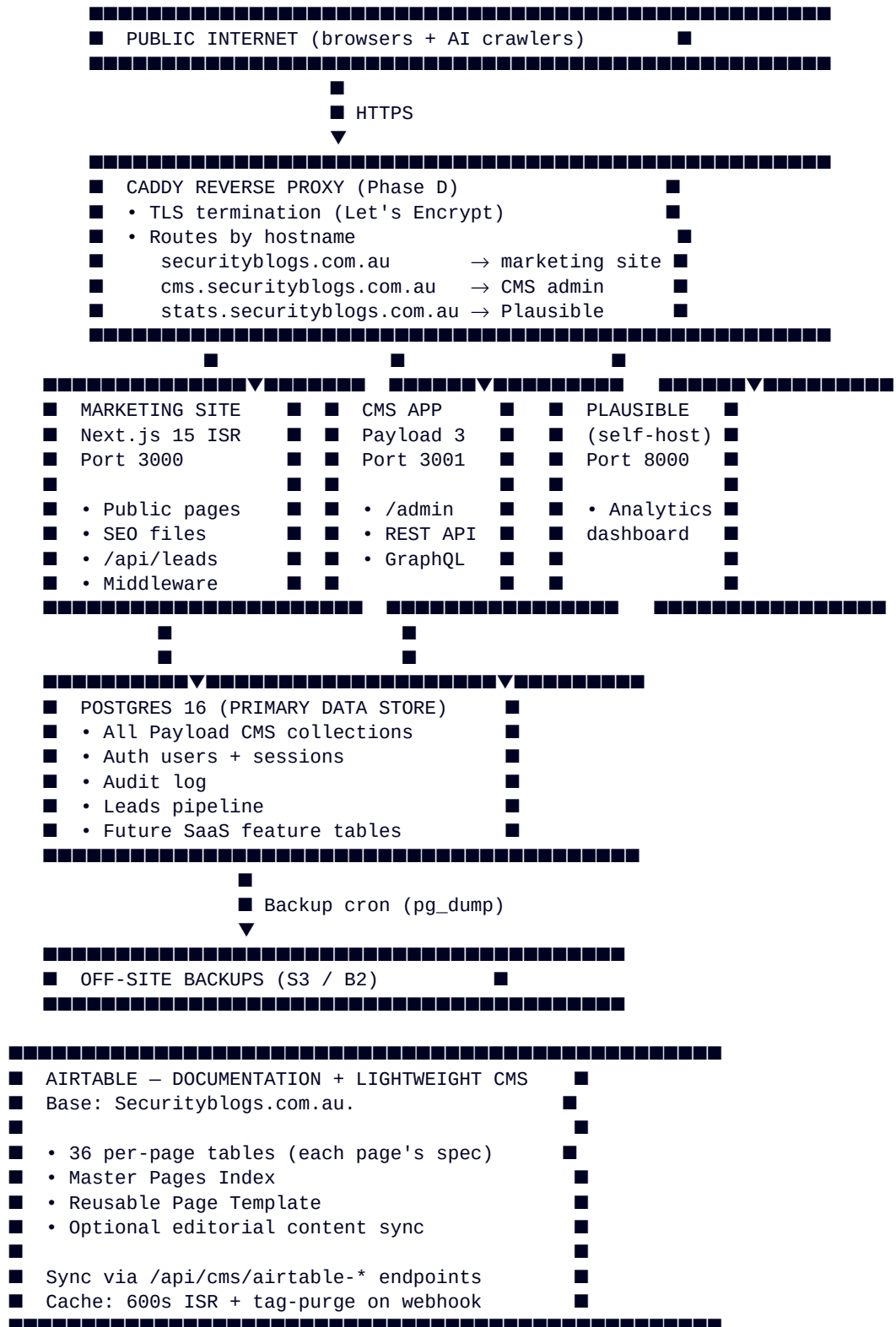


SYSTEM_ARCHITECTURE

Repository: github.com/Jonaid880/Security-Blogs **Architecture model:** Hybrid (Postgres-primary, Airtable as documentation + lightweight content layer) **Date:** 2026-06-14

1. Topology



2. Component inventory

Layer	Component	Purpose	Source of truth
Frontend	Next.js 15 marketing site	Public-facing website	Code in <code>app/</code> , <code>components/</code> , <code>lib/</code>
Frontend	React 19 + Tailwind 4	UI rendering	
Frontend	Lucide React, Framer Motion, Mapbox GL	Visual elements	
Backend	Payload CMS 3	Editorial admin + REST/GraphQL API	<code>cms/</code> directory
Backend	Next.js API routes	Form submission, CMS aggregator, redirect hit-counter	<code>app/api/</code>
Backend	Middleware (Edge)	CMS-driven redirects, rate limit, ISR cache	<code>middleware.ts</code>
Data	Postgres 16	All transactional + content data	<code>cms/</code> schemas auto-migrate
Data	Airtable (app4m600zymaqPKHX)	Documentation, page-by-page implementation specs	38 tables today
Auth	Payload built-in auth (Argon2id + JWT)	Editorial users	<code>cms/src/collections/Users.ts</code>
Auth	NextAuth (future)	End-user / SaaS auth	Phase E
Email	Hostinger SMTP via nodemailer	Lead notifications	<code>cms/src/email/transport.ts</code>
Email	MailHog (dev)	Catches mail locally	<code>docker-compose.yml</code>
Edge	Cloudflare Turnstile	Anti-bot on forms	<code>components/Turnstile.tsx</code>
Edge	Cloudflare (proposed)	CDN + WAF in Phase D	TBD
Analytics	GTM (current) + Plausible (planned)	Site analytics	<code>Settings.analytics.gtmId</code>
Maps	Mapbox + Google Maps	GEO map + local SEO widget	<code>lib/env.ts</code> (referrer-restricted)
Search	Ahrefs Brand Radar MCP	AI visibility scoring (future)	Server-side only

3. Data flow patterns

3.1 Public page request (ISR-cached)

Browser

- Cloudflare CDN (Phase D)
- Caddy
- Next.js server (marketing site, port 3000)
- `middleware.ts` checks CMS redirect rules (60s cache)
- `page.tsx` async server component

- lib/cms.ts → Payload REST /api/<collection> (60-600s cache)
- Postgres SELECT via Payload ORM
- renders HTML
- Cloudflare CDN caches
- returns to browser

3.2 Form submission (lead capture)

- Browser POST /api/leads with JSON body + Turnstile token
- middleware.ts (no redirect match, passes through)
 - app/api/leads/route.ts handler
 - Honeypot check (silent OK if bot)
 - Cloudflare Turnstile siteverify (server-to-server)
 - Per-IP rate limit (15 req/15 min, in-memory)
 - Shape validation (zod-equivalent inline)
 - lib/cms.ts createLead() → POST /api/leads on CMS (with PAYLOAD_API_KEY)
 - Payload validates + writes to Postgres `leads` table
 - afterChange hook sends email to assignedTo via Hostinger SMTP
 - return { ok: true, id: leadId } to browser

3.3 Editorial change

- Editor logs into /admin (CMS port 3001)
- Payload checks Argon2id hash + issues JWT + sets cookie
 - Edits a Page record's blocks
 - Autosave fires every 2s → POST /api/pages/:id (draft)
 - On Publish → status=published, publishedAt=now (hook)
 - Optional: afterChange hook calls /api/cms/purge?tag=page:home on marketing site
 - Marketing site revalidates next request

3.4 Documentation update

- Engineer edits Documentation/Markdown/PAGE_X.md
- Commits to git
 - Pushes to phase-c-frontend-rewire branch
 - GitHub Action (planned) runs convert_md_to_pdf.py
 - PDFs appear in Documentation/PDFs/
 - Optional: webhook updates Airtable record with PDF attachment URL

4. Trust boundaries

Boundary	What crosses it	Validation
Browser → Marketing site	Form submissions, page requests	Turnstile, honeypot, rate limit, shape validation
Marketing site → CMS	Server-to-server API calls with PAYLOAD_API_KEY	TLS, API key auth, Payload access policies
CMS → Postgres	Direct DB connection	Connection string with credentials in env
Marketing site → Airtable	Documentation sync	PAT token, server-side only
Admin → CMS UI	Editorial work	Argon2id + JWT + role-based access
Hostinger SMTP → Inbox	Outbound mail	SPF / DKIM / DMARC (Phase D)
MCP tool calls (AI scoring)	API key passed server-side	Never exposed to browser

5. Hostnames and ports

Hostname (prod)	Internal port	Purpose
securityblogs.com.au	3000 (internal)	Marketing site
www.securityblogs.com.au	3000	Redirects to apex
cms.securityblogs.com.au	3001 (internal)	Payload admin + API
stats.securityblogs.com.au	8000 (internal)	Plausible self-host
(internal)	5432	Postgres (not exposed externally)
(dev)	1025 / 8025	MailHog (dev only)

6. Environment-by-environment matrix

Aspect	Local dev	Staging (proposed)	Production (Phase D)
Marketing site URL	http://localhost:3000	https://staging.securityblogs.com.au	https://securityblogs.com.au
CMS URL	http://localhost:3001	https://cms.staging.securityblogs.com.au	https://cms.securityblogs.com.au
Postgres	Docker container db	Managed Postgres (Hostinger / Neon)	Managed Postgres
Email	MailHog at localhost:8025	MailHog or test inbox	Hostinger SMTP (info@)
Turnstile	Empty key → skipped	Test site keys	Real site keys
ISR revalidate	0 (off) or 60s	60s	60–600s by tag
Sentry / monitoring	Off	On (test DSN)	On (prod DSN)
Plausible	Off	On (subdomain)	On (stats.)

7. Repository layout

Security-Blogs/	← repo root
■■■■ app/	← FRONTEND – Next.js App Router pages
■ ■■■■ api/	
■ ■■■■ leads/route.ts	← form submission endpoint
■ ■■■■ services/[slug]/page.tsx	← dynamic service page (CMS)
■ ■■■■ case-studies/[slug]/page.tsx	← dynamic case study (CMS)
■ ■■■■ knowledge-hub/[category]/	
■ ■■■■ [slug]/page.tsx	← dynamic blog post (CMS)
■ ■■■■ sitemap.ts	← CMS-aware sitemap
■ ■■■■ robots.ts	← CMS-aware robots
■ ■■■■ llms.txt/route.ts	← CMS-aware llms.txt
■ ■■■■ [36 hardcoded legacy pages]	← migrated in Phase C.2
■■■■ components/	← FRONTEND – React components
■ ■■■■ layout/Navbar.tsx	
■ ■■■■ layout/Footer.tsx	
■ ■■■■ modules/ModuleRenderer.tsx	← block-based page renderer
■ ■■■■ modules/blocks/*.tsx	← 9 block components
■ ■■■■ modules/LexicalRenderer.tsx	← Lexical JSON → JSX walker
■ ■■■■ Turnstile.tsx	← Cloudflare widget
■ ■■■■ [...service-page previews, hero variants...]	
■■■■ lib/	← FRONTEND – utilities
■ ■■■■ cms.ts	← typed Payload REST client
■ ■■■■ cmsTypes.ts	← read-shape types
■ ■■■■ submitForm.ts	← form helper (calls /api/leads)

■ ■■■ site.ts	← hardcoded constants (Phase C.2 deletes)
■ ■■■ env.ts	← public env wrappers
■■■ middleware.ts	← Edge middleware (CMS redirects)
■■■ next.config.mjs	← ISR config; output:'export' removed
■■■ package.json	← marketing site deps
■	
■■■ cms/	← BACKEND – Payload CMS app
■ ■■■ app/(payload)/	← Next.js routes hosting Payload
■ ■■■ src/	
■ ■ ■■■ collections/	← 9 collections
■ ■ ■■■ globals/Settings.ts	← 1 global
■ ■ ■■■ access/	← role helpers
■ ■ ■■■ lib/aiVisibilityFields.ts	← shared field group
■ ■ ■■■ email/transport.ts	← nodemailer adapter
■ ■ ■■■ seed/	← 8 idempotent seeds
■ ■■■ payload.config.ts	← central Payload config
■ ■■■ package.json	← CMS deps
■	
■■■ docker-compose.yml	← Postgres 16 + MailHog
■■■ docs/	← Phase-based engineering docs
■ ■■■ 01-architecture.md	
■ ■■■ 02-installation-local.md	
■ ■■■ 03-env-variables.md	
■ ■■■ 04-admin-portal-guide.md	
■ ■■■ 05-phases.md	
■ ■■■ 06-frontend-data-flow.md	
■	
■■■ Documentation/	← THIS deliverable
■ ■■■ Markdown/	← .md sources
■ ■■■ PDFs/	← generated .pdf
■ ■■■ convert_md_to_pdf.py	← converter
■ ■■■ DOCUMENTATION_INDEX.csv	← Airtable import
■	
■■■ BACKEND_AUDIT_REPORT.md	← static analysis snapshot
■■■ AIRTABLE_SCHEMA_MAP.md	← if Airtable were primary CMS
■■■ COMPLETE_SYSTEM_ARCHITECTURE.md	← (this deliverable in MD form)
■■■ public/	← static assets (favicon, logos)

8. Production-readiness rollup

System	Current %	Production-ready	Critical blockers
Frontend (visual)	100	Yes	None
Frontend (CMS integration)	15	No	Phase C.2 per-page migration
SEO surface	100	Yes	None
Static deployment	100	Yes (current) / No (this branch)	Branch broke static export; needs Phase D VPS
Node.js backend (CMS)	85	No	Never booted; needs runtime validation
Postgres schema	80	No	First migrate not run
Admin panel	95	No	Boot not verified
Production server	0	No	Whole Phase D pending
Dynamic content	25	No	CMS not yet authoritative for pages

System	Current %	Production-ready	Critical blockers
Future SaaS platform	0	No	Not started; auth + billing + tenant model TBD

9. Recommended next actions

1. Validate Phase A + B + C.1 locally (5h). Boot Docker, migrate, seed, walk admin.
2. Phase C.2 per-page migration (30–45h). Home → about → contact → services delete → knowledge-hub category landings.
3. Multipart `/api/leads/upload` for CV (4–6h).
4. CMS-side `/api/redirects/:id/hit` companion endpoint (2h).
5. Phase D infrastructure (40–60h). VPS, Caddy, systemd, backups, Plausible, secrets vault.
6. Tests (16–24h). Contract tests on `lib/cms.ts`, Playwright happy-path.
7. Cut-over (4–6h). DNS TTL drop, deploy, DNS swap, 24h watch.

Total: **~95–140 hours** to production-ready.

End of SYSTEM_ARCHITECTURE.md